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A demonstration of a network isolation solution for the computing continuum

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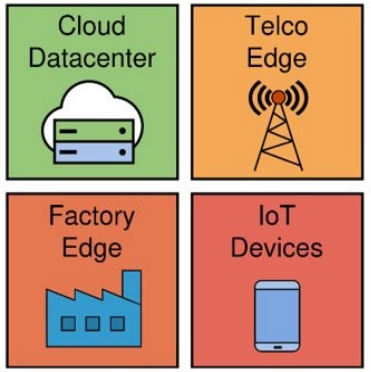
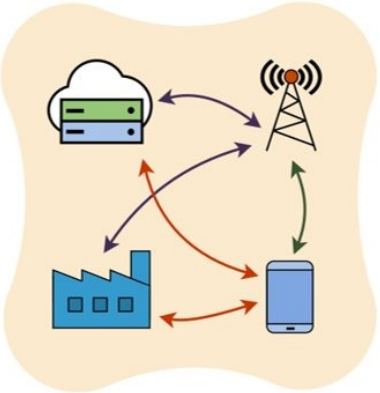
State of the Art and Motivation

- **Computing continuum** is a novel cloud paradigm aiming to unify heterogeneous infrastructure layers (e.g., cloud, edge, IoT) creating a seamless pool of resources shared among multiple users.
- **Physical boundaries are replaced by a dynamic and virtual border** spanning across the resources of the continuum.
- Several challenges:
  - *extreme heterogeneity of the underlying components*
  - *multi-cluster environment spanning diverse layers*
  - *multiple actors with distinct and potentially conflicting goals*

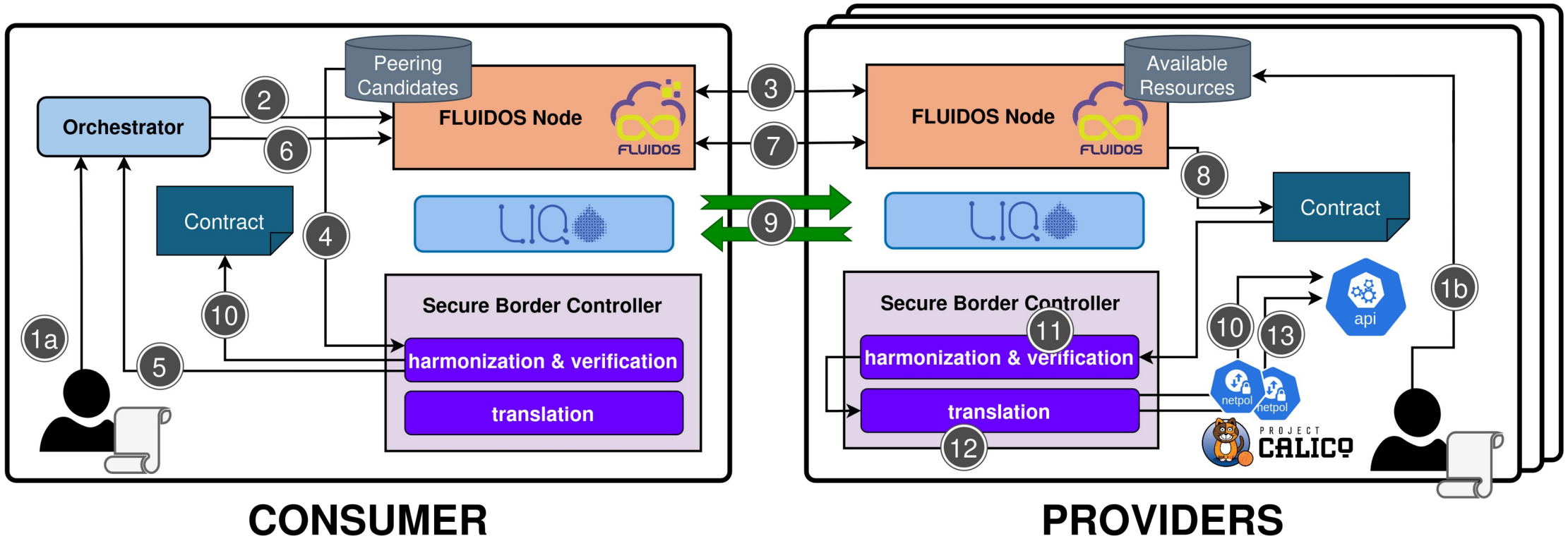


(a) isolated clouds


(b) unified continuum

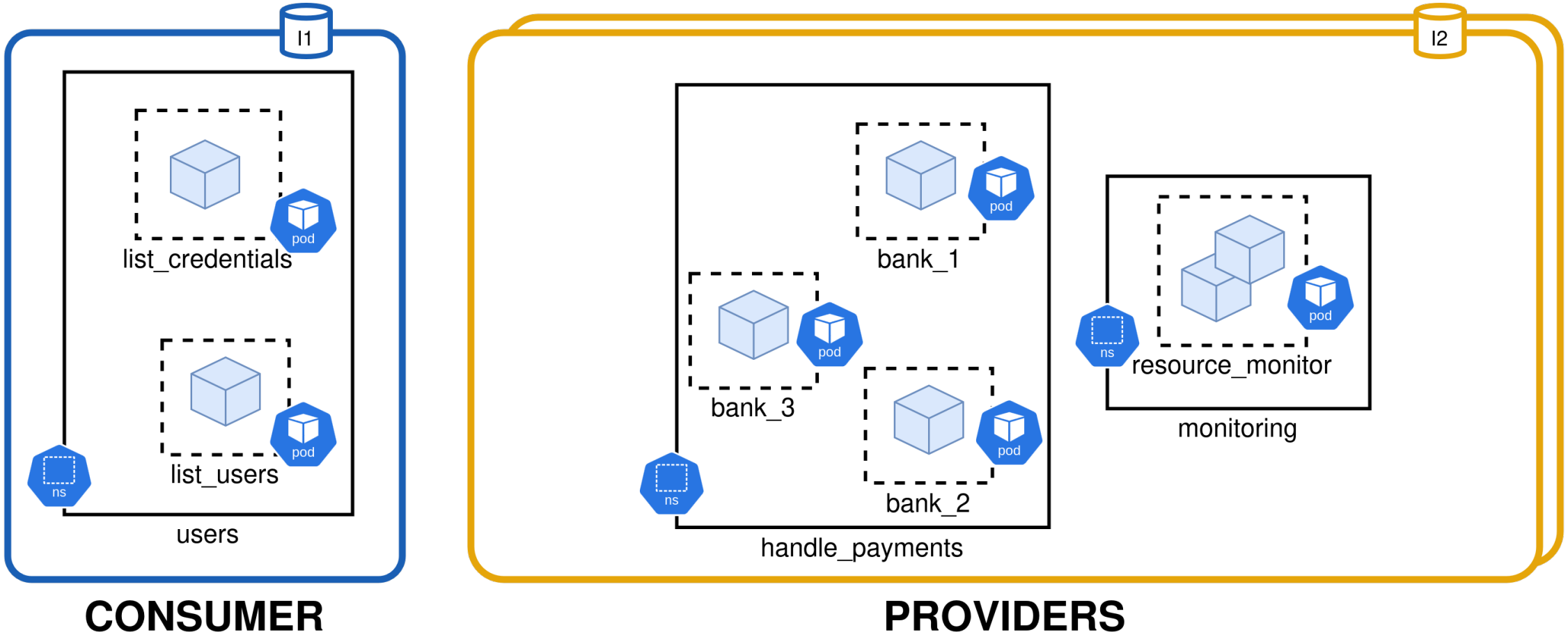
No current approach can provide network isolation for the computing continuum scenario.

The proposed solution: Intent-based border protection



The proposed solution consists in an **autonomous** framework to instantiate a **secure border** through the configuration of network isolation primitives across the computing continuum through **user-defined intents**.

Example scenario



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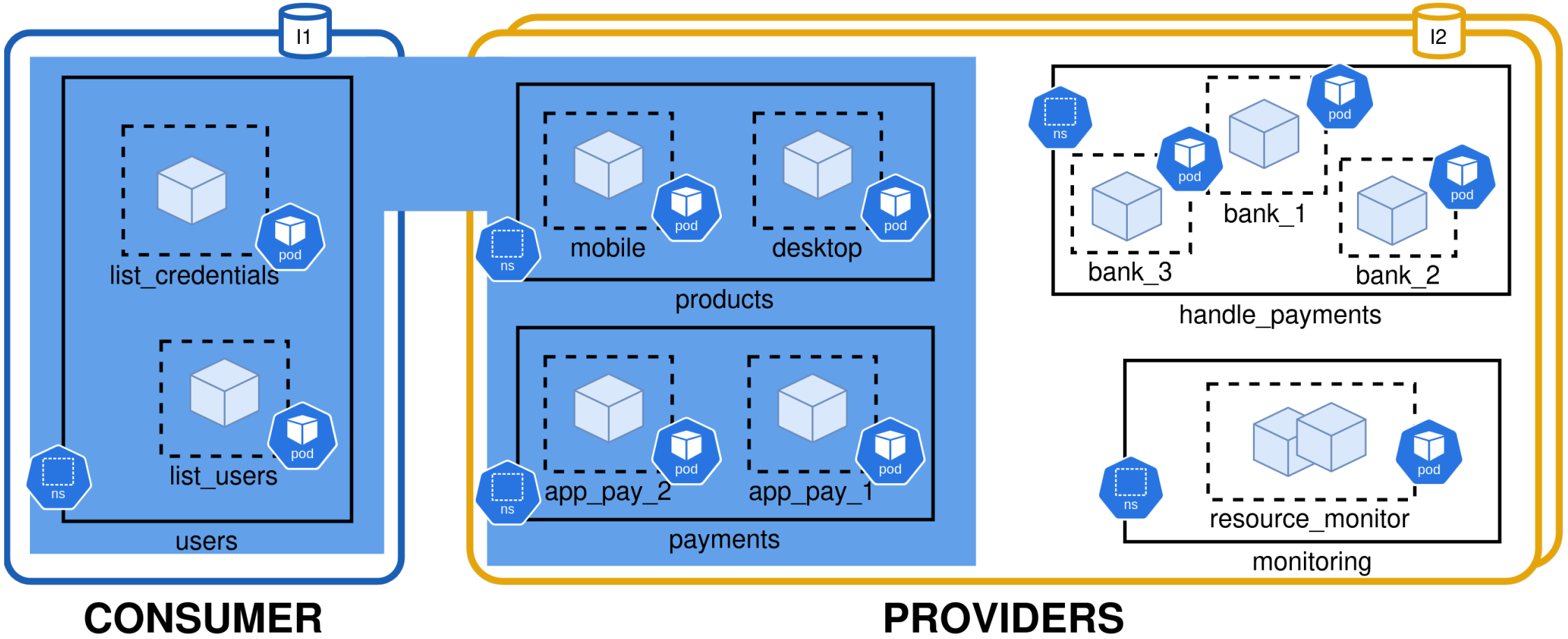

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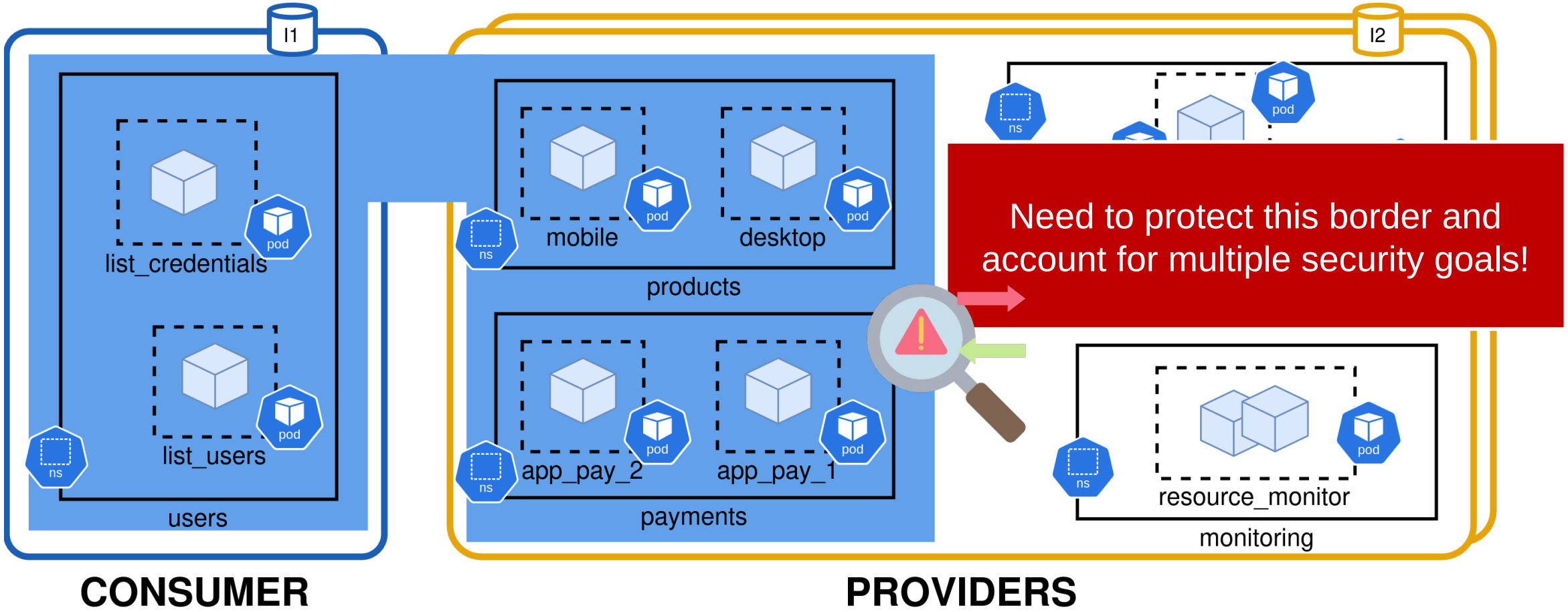
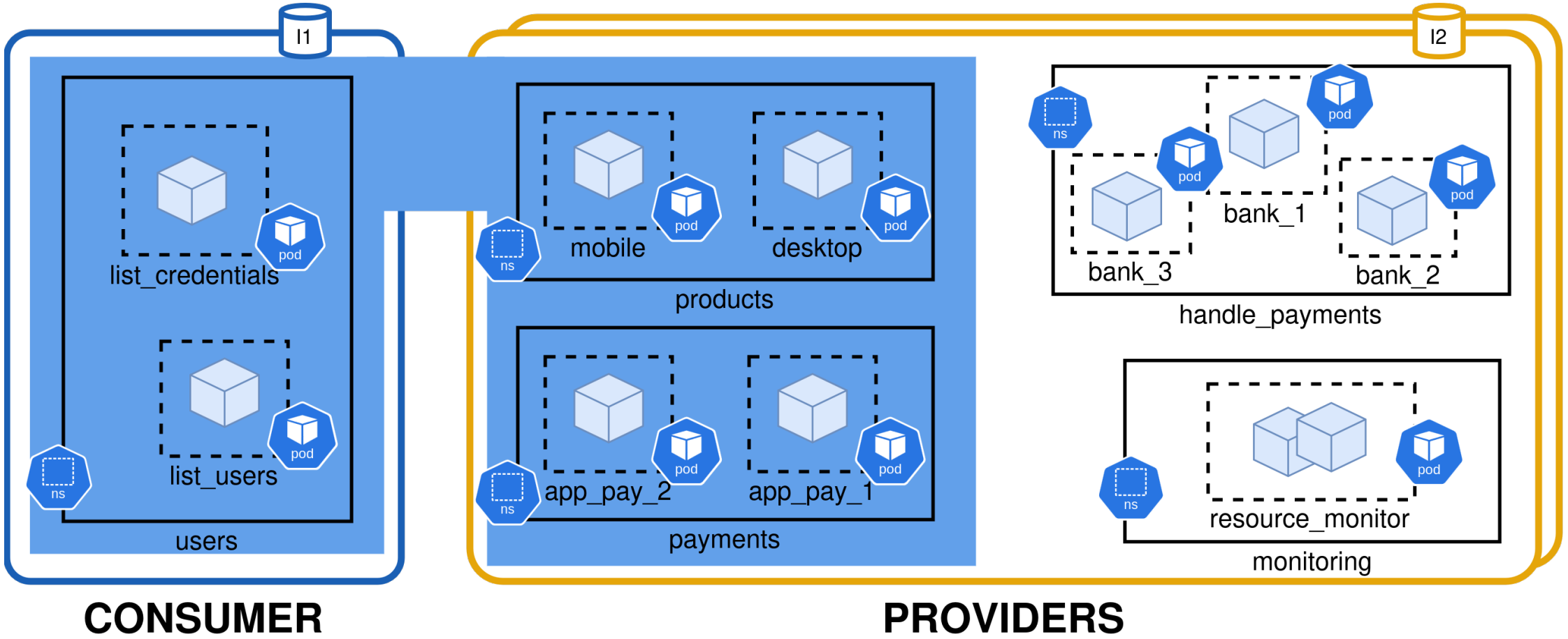


Example scenario: Request Intents



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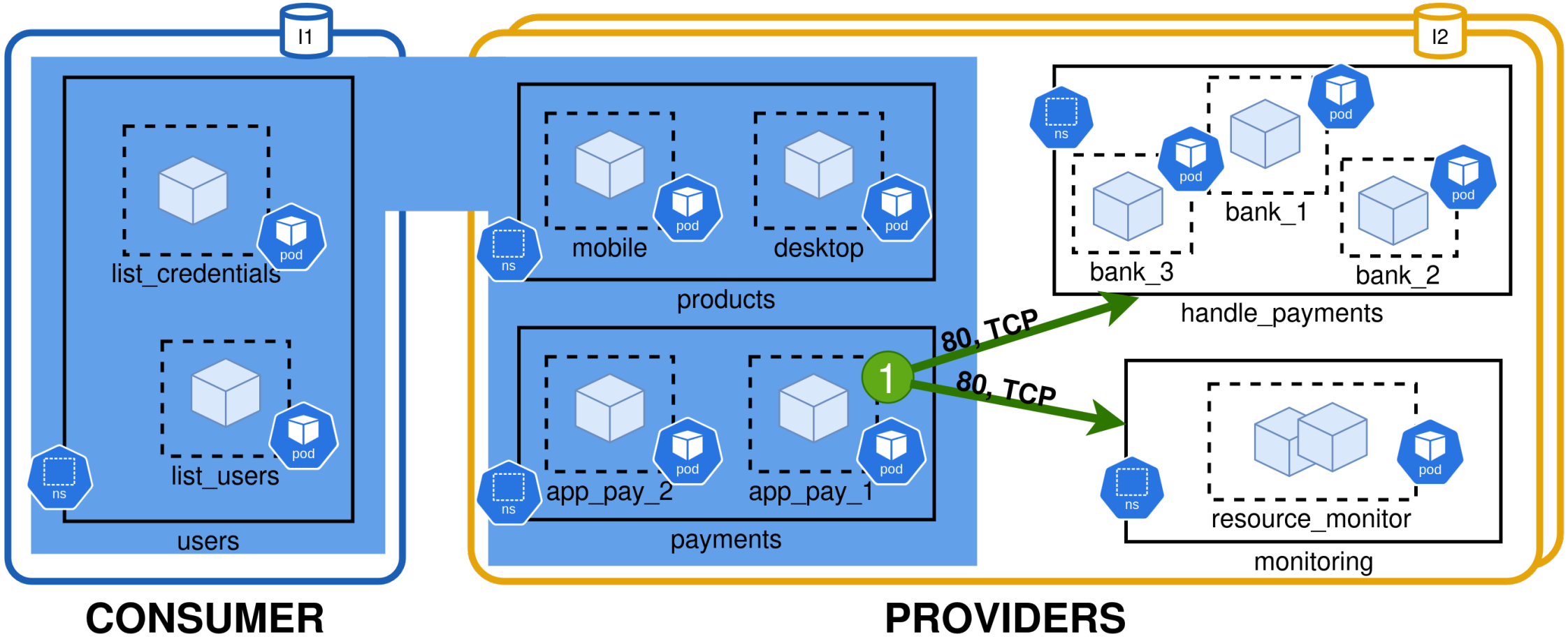

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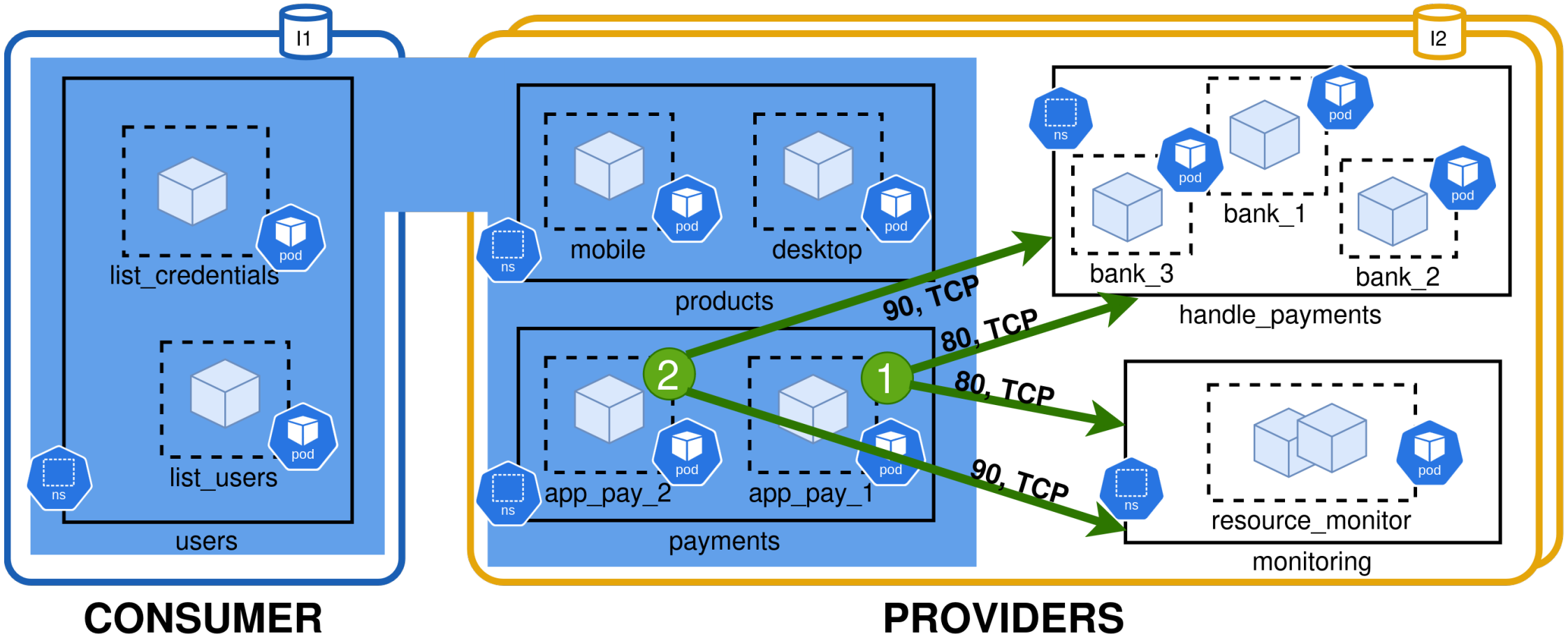

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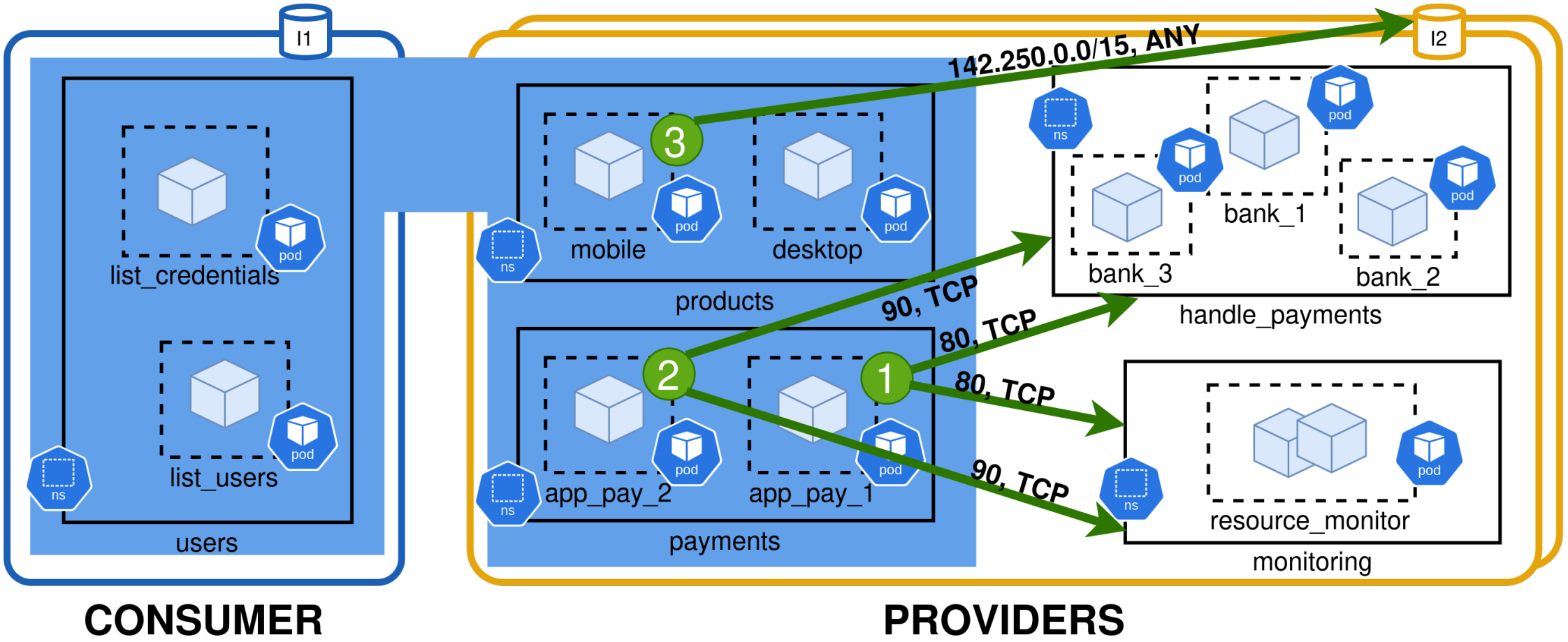

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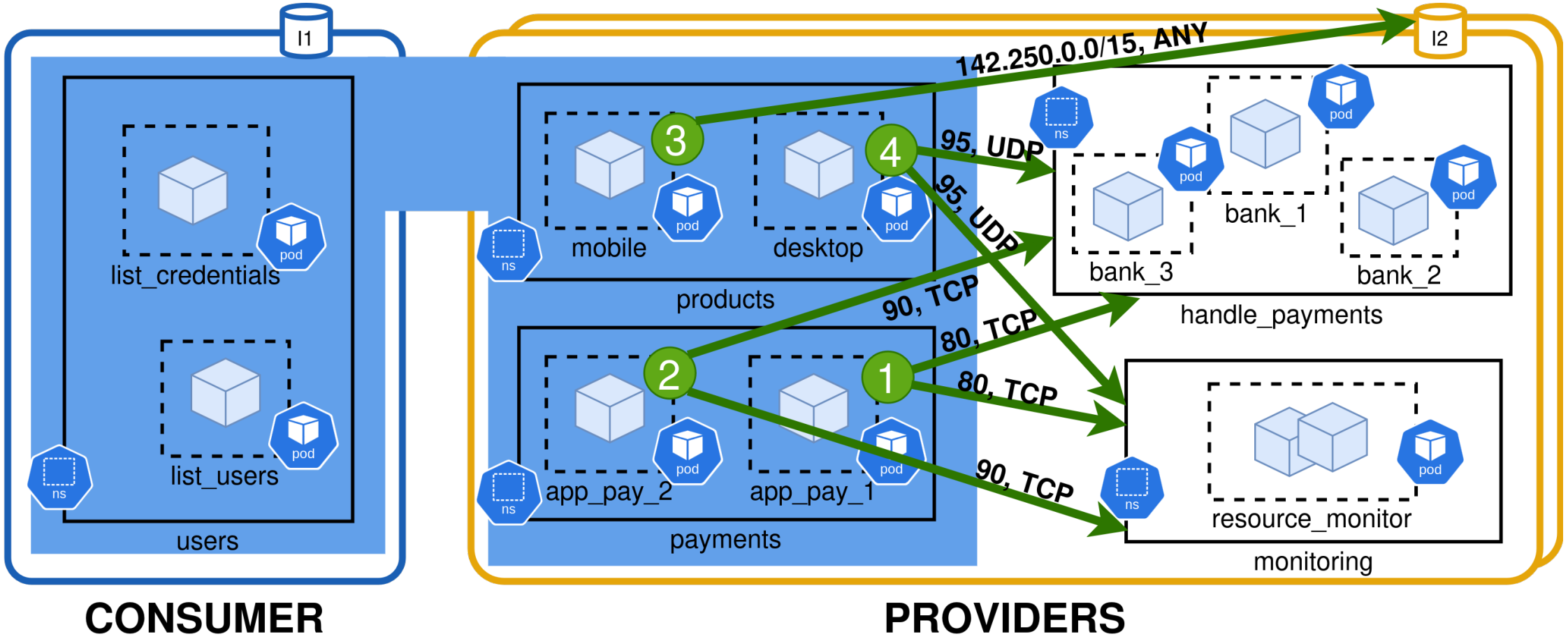
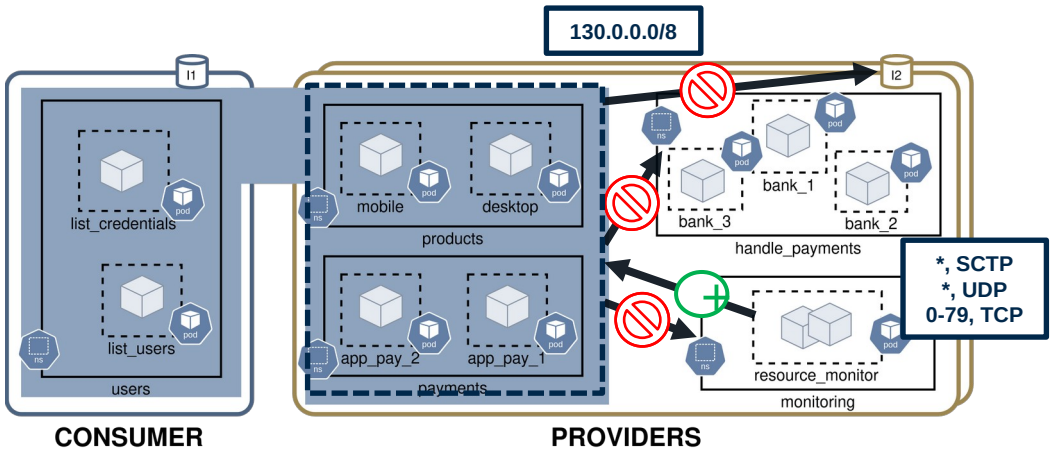
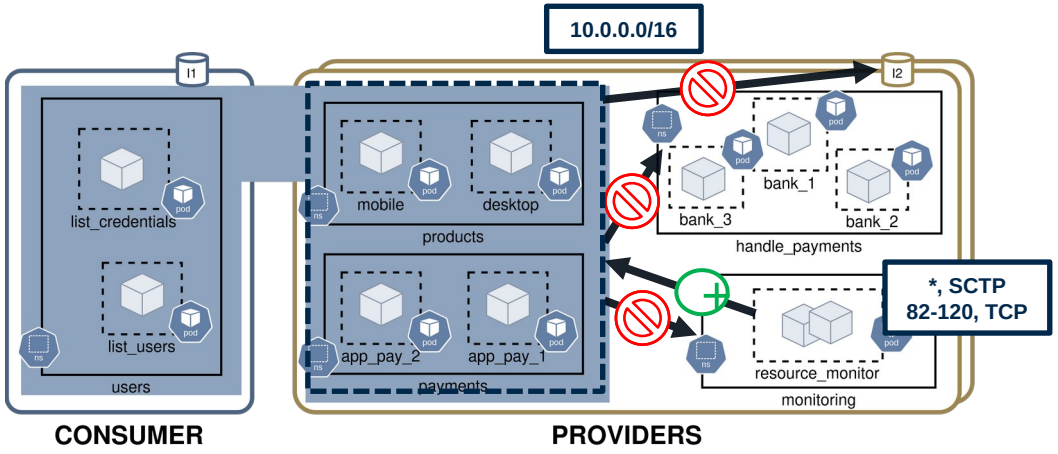
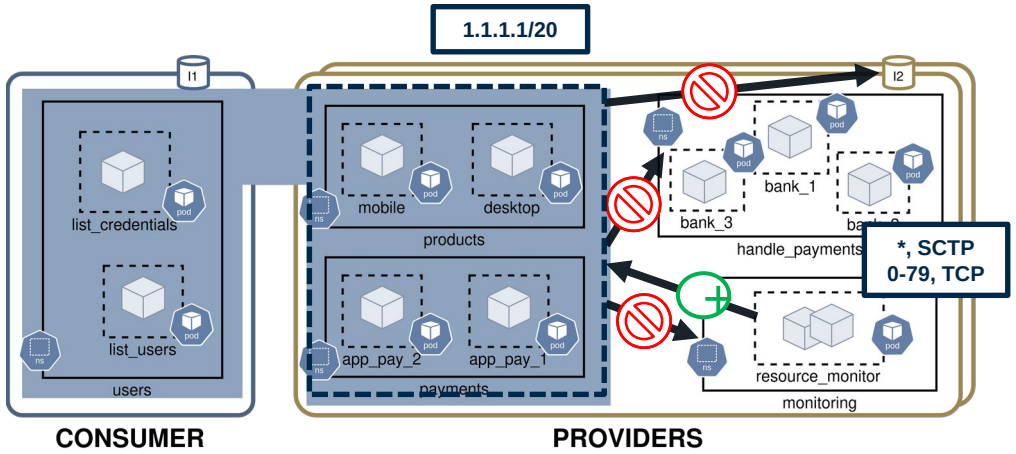
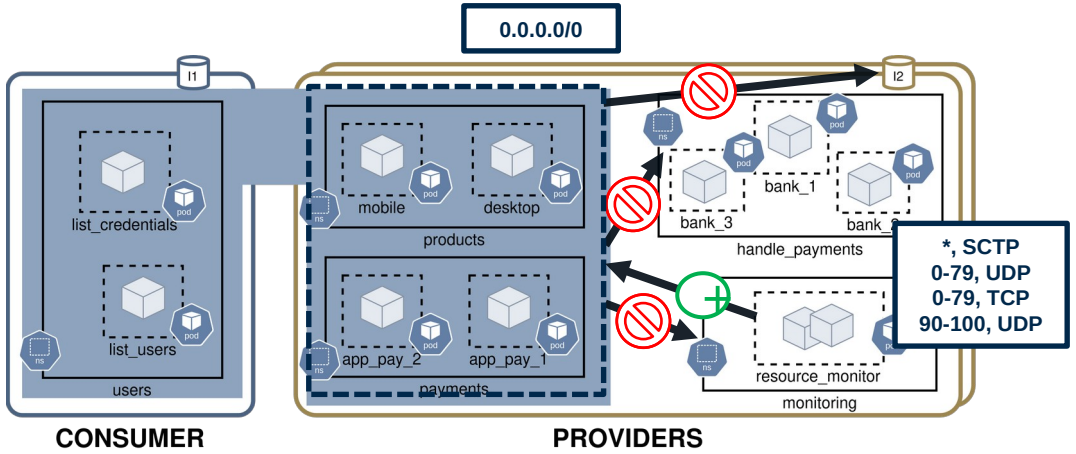


Example scenario: Authorization Intents









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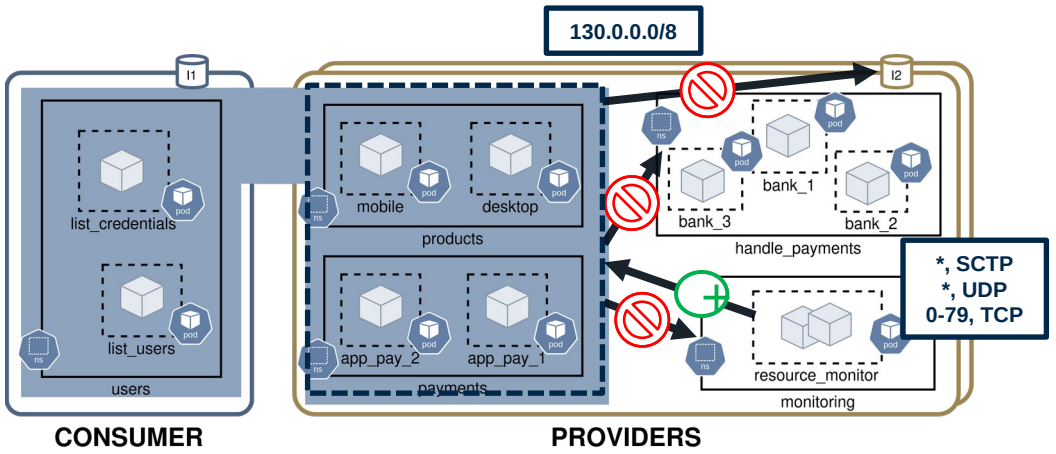

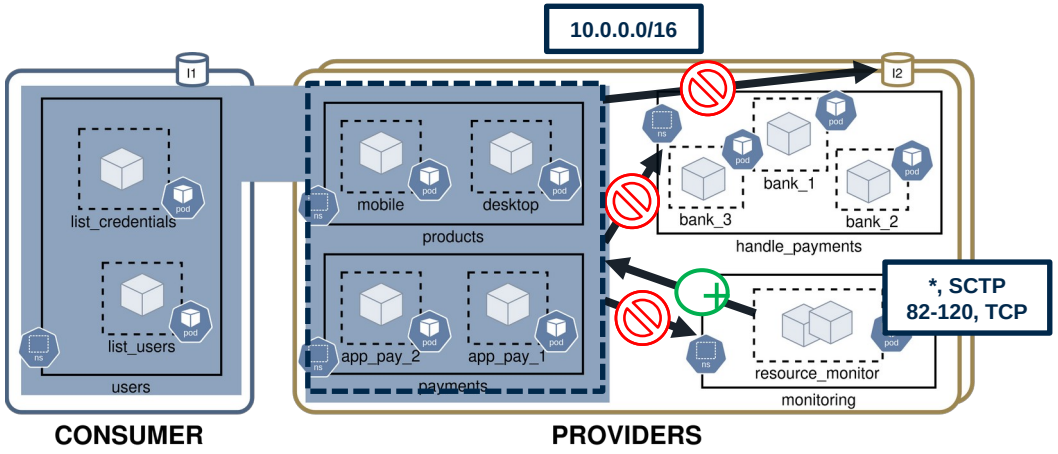

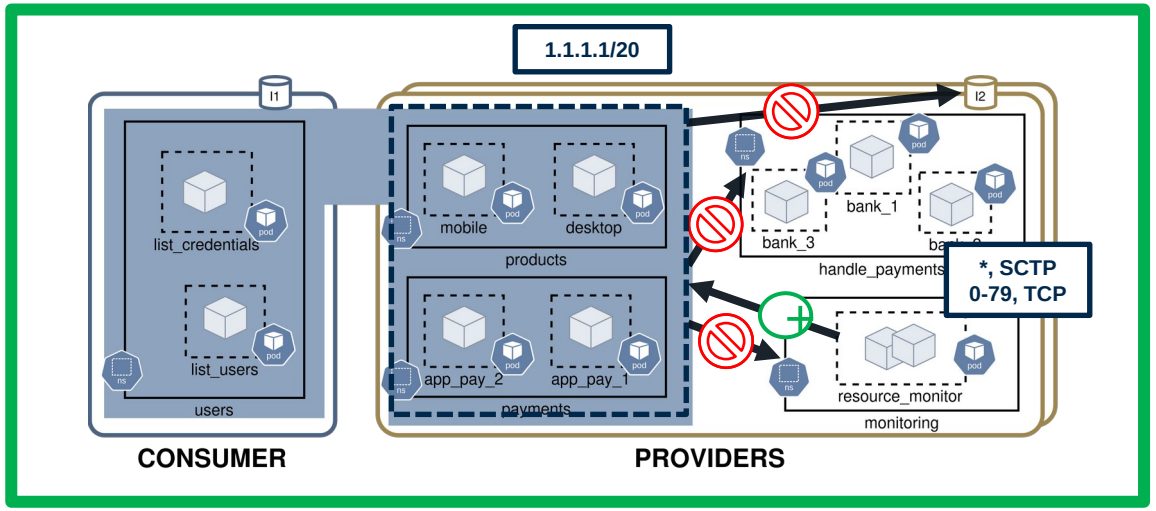

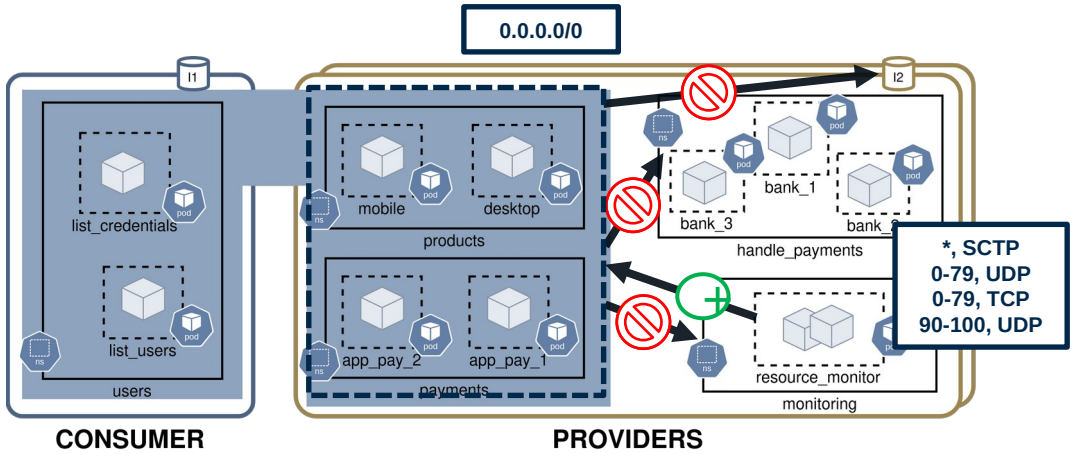

Thanks for your attention!

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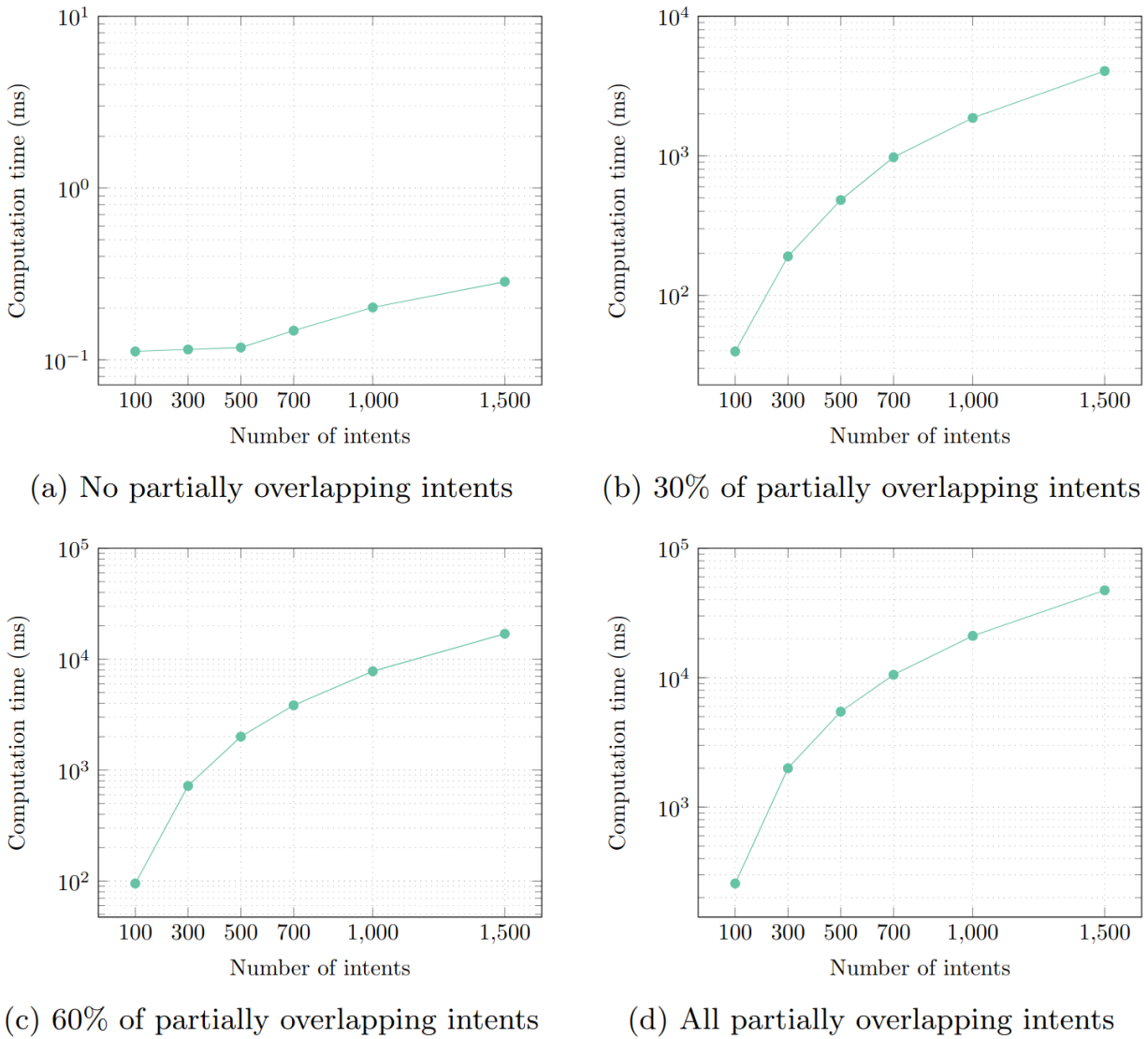
GitHub:

F. Pizzato, D. Bringhenti, R. Sisto, F. Valenza, "*An intent-based solution for network isolation in Kubernetes*". IEEE Conf. on Network Softwarization (NetSoft), Saint Louis, USA, 24-28 June 2024. 10.1109/NetSoft60951.2024.10588939

F. Pizzato, D. Bringhenti, R. Sisto, F. Valenza, "*Intent-driven network isolation for the cloud computing continuum*", Journal of Network and System Management. https://doi.org/10.1007/s10922-025-09986-1

Scalability Analysis of Harmonization phase



(a) No partially overlapping intents

(b) 30% of partially overlapping intents

(c) 60% of partially overlapping intents

(d) All partially overlapping intents

Fig. 13: Scalability of Harmonize function to varying size and complexity

Scalability Analysis of Verification and Translation phases

Test scenario description	Avg. Computation Time (ms)
Incompatible parameter (accept monitoring)	0.0030435
Single non-overlapping field (protocol)	0.0403948
Single non-overlapping field (port)	0.0414726
Single non-overlapping field (src, with S_{CIDR})	0.0393982
Single non-overlapping field (src, with S_{PodNs})	0.0313571
Single non-overlapping field (dst, with S_{CIDR})	0.0428547
Single non-overlapping field (dst, with S_{PodNs})	0.0280475
Single non-overlapping field (with different selector types)	0.0274011
All overlapping fields (srd and dst with S_{CIDR})	0.0358931
All overlapping fields (srd and dst with S_{PodNs})	0.0266328
All overlapping fields (srd and dst with mixed selectors)	0.0349272

Table 5: Performance analysis of Verification function with varying complexity.

Test scenario description	Avg. Computation Time (ms)
Exclusive usage of S_{CIDR} for both src and dst	0.0030435
Exclusive usage of S_{PodNs} for both src and dst	0.0403948
Mixed usage of S_{CIDR} and S_{PodNs} for src and dst	0.0414726

Table 6: Performance analysis of Translation function.

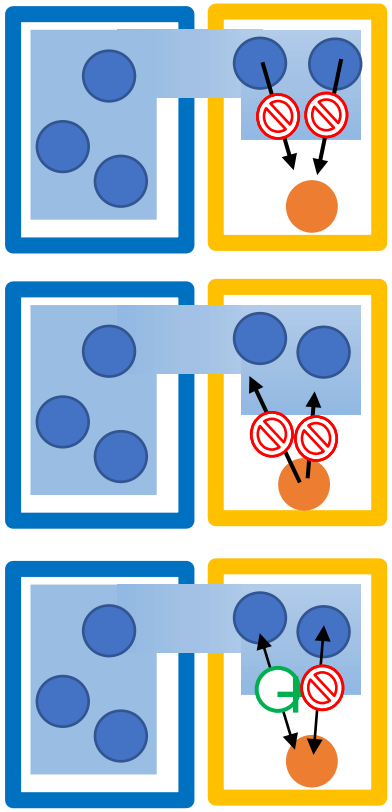
Network isolation

A significant problem is **network isolation**, which involves protecting the dynamic virtual boundaries extending over different physical machines and administrative domains. It is a multi-purpose problem:

1. *Protect the hosting cluster against potential harm from guest applications deployed by a different tenant*

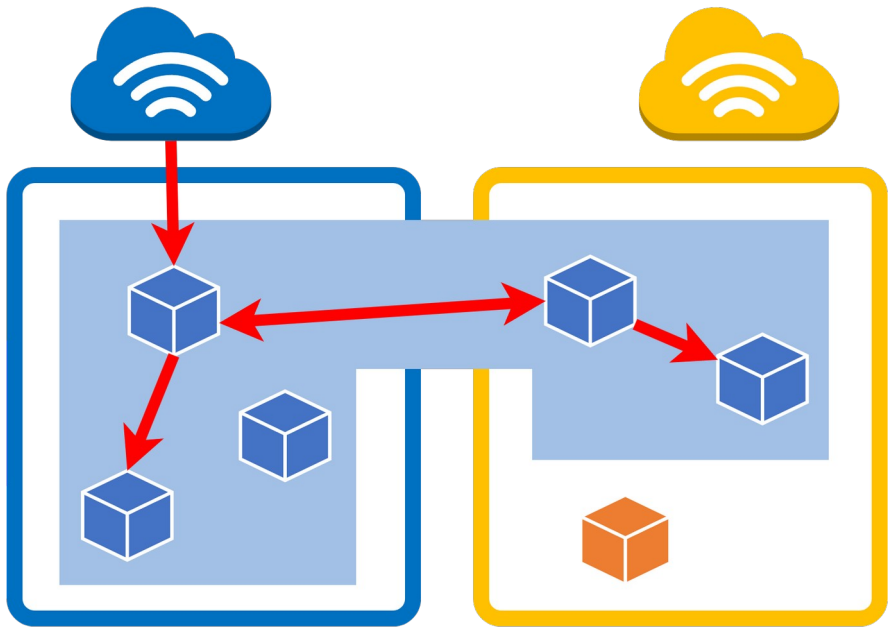
2. *Protect guest applications against potential stealing of data or code or unauthorized interference from the host*

3. *Allow application owners to precisely specify which interactions their applications may have with the environment*



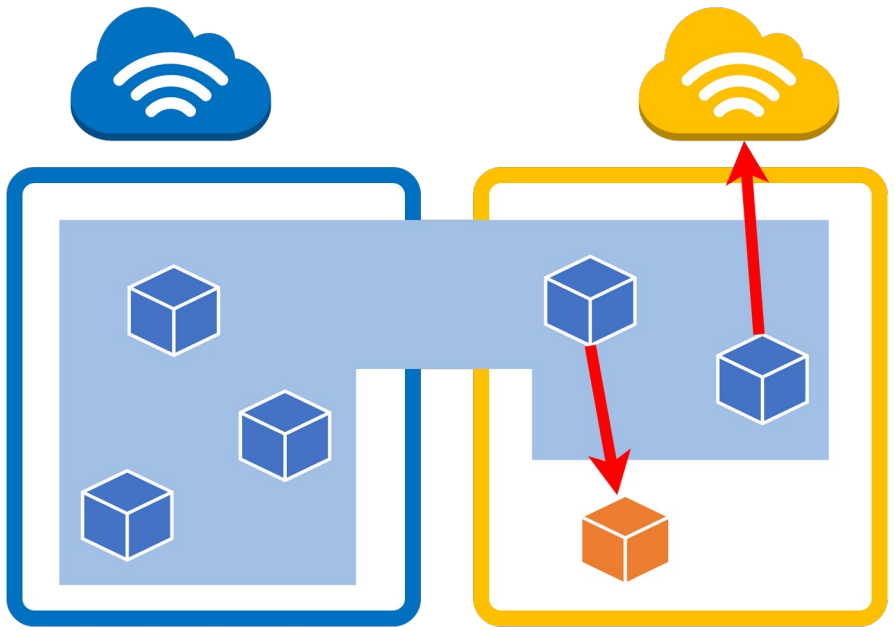
Private Intents

- They are related to communications happening within the virtual cluster (i.e., intra-virtual cluster), involving both the local and remote resources.
- These are <u>not</u> subject to the authorization of the host(s). The principle is that each user should have unlimited control over his own resources



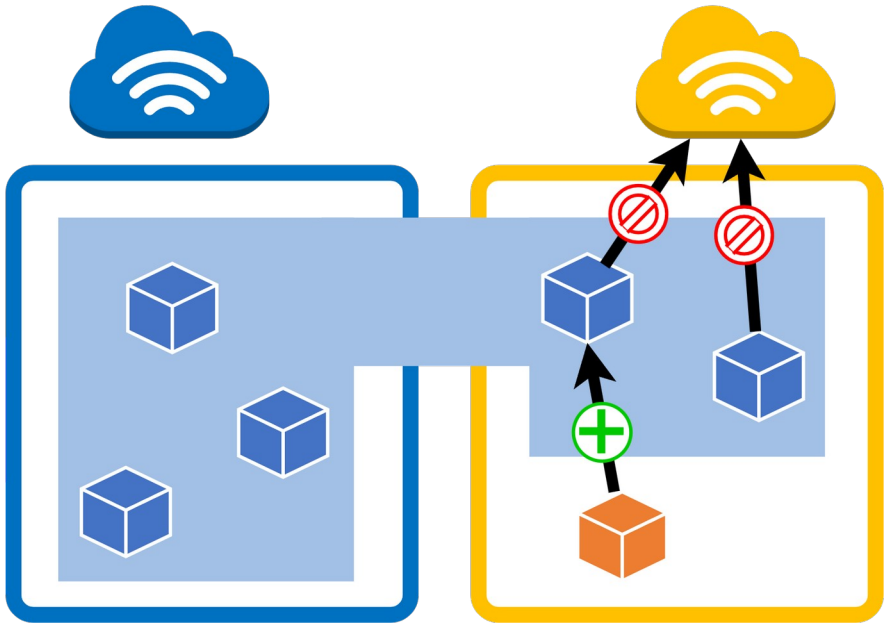
Request Intents

- They are related to communications crossing the virtual border of an extended cluster and could be configured to target services offered by the hosting cluster or addresses on the external network.
- The consumer could tune some parameters to refine his will in terms of isolation with respect to the hosting cluster (e.g., to accept or refuse to be monitored by the provider)



Authorization Intents

- They represent the authorization policies defined by a user with respect to any possible guest, to regulate the inter-virtual cluster communications of each hosted cluster.
- They target two communication types:
  - **DeniedCommunications** must always be blocked or filtered for all hosts.
  - **MandatoryCommunications** must be injected to all hosts (e.g., to whitelist some monitoring probes)



Intents' Format

"from SRC to DST, protocol[:port[-endPort]]"

- *SRC* and *DST* can be either a Pod, or a group of Pods, with the same label and an associated Namespace, or an CIDR address. Could use wildcard.
- *Protocol* can be any transport protocol (TCP, UDP, STCP, etc.) or the value ALL to represent al of them.
- *Port* can be a port or a range of ports.

- Request and Private intents can be expressed only in whitelisting, so as to be compliant with the default behavior of Kubernetes Network Policy.
- Authorization intents allow for more expressiveness. The *DeniedCommunications* are expressed in blacklisting, and conversely the *MandatoryCommunications* are simply allowed communications.